Operational Research (2022) 22:2973–3001
https://doi.org/10.1007/s12351-021-00640-7

ORIGINAL PAPER

An optimization approach to design forest road networks and plan timber transportation

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Received: 30 November 2020 / Revised: 19 March 2021 / Accepted: 13 April 2021 /
Published online: 24 April 2021
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Abstract
The provision of ecosystem services depends on the landscape-level distribution of forest management options over extended planning horizons. Nevertheless, transportation costs are an important fraction of total procurement costs. Therefore, forest managers are challenged by the need to make decisions regarding the design of the road network needed to have access to forest management units (e.g., stands) as well as the transportation of forest products over this network. The objective of this research is to develop an approach to integrate the design and maintenance of a forest roads' network and to provide solutions to the timber loading location site and the timber transportation problems. The emphasis is on the combination of mathematical programming formulations and heuristic solution techniques to minimize the sum of road building, road maintenance, timber loading and transportation costs. Specifically, a mixed integer programming model is built to represent the problem while a heuristic algorithm, based on a Graph Simplification procedure and a big-M Estimation procedure, is proposed to solve the underlying optimization problem. Results from an application to a large-scale forest management problem in Northwest Portugal involving 1137 management units are discussed. Results suggest that this approach may help address efficiently the need by forest managers to integrate road design, road maintenance, timber loading location and transportation concerns.

Keywords Forest road network design · Road building · Road maintenance · Timber transportation · Mixed integer linear programming model · Big-M estimation procedure

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1 Introduction

One of the main challenges to forest management planning relies on the design of a forest road network that may support the optimization of the transportation of timber from stands that have been harvested to existing roads. The literature reports the research by several authors to address this challenge (e.g. Weintraub and Navon 1976 and recently Naderializadeh and Crowe (2018), Ross et al. (2018)). The problem is framed by spatial issues, e.g., by the need to design the forest road network in concordance with harvesting decisions, namely by providing access of machinery to the stands to be harvested. It may be framed further by temporal considerations, e.g., by the need to integrate operational, tactical and long-term decisions. Nevertheless, this integration may increase the computational complexity of the solution approaches. Thus, according to the combinatorial nature and size of real-life cases under study, operational, tactical and long-term optimization problems may be addressed separately or in an integrated mode.

Some simplifications are usually assumed while considering the integrated harvesting scheduling and road network design problem, which may be restrictions to small forest areas, special network road configurations or short temporal horizons. Weintraub and Navon (1976), in their seminal paper presented a Mixed Integer Linear programming (MILP) model with binary variables associated to decisions of building road segments, whilst forest management and traffic flow were modeled through continuous variables. Afterwards, other authors focused their research on the integrated harvesting, road building and timber transportation problem (e.g., Jones et al. (1991); Guignard et al. (1998); Palma and Nelson (2014)) by developing different solution approaches to address real life instances of forest management problems. Andalaft et Al. (2003) also considered the possibility of road upgrading and proposed a solution approach based on model strengthening, lifting constraints and Lagrangean relaxation. Recently, Naderializadeh and Crowe (2018) and Ross et al (2018) introduced some new features in the underlying road network subproblem. Naderializadeh and Crowe (2018) observed that timber transported along roads always flows in the same direction, so they propose a MILP model in which road segments are directed. They compared their formulation with others where the road segments are undirected. For two real instances with 707 and 900 stands and 2877 and 3418 candidate roads, respectively, the solver CPLEX obtained solutions with gaps outperforming the other approaches considered. Ross et al (2018) assumed the road network already exists, but roads must be reconstructed, the cost being proportional to the number of years since the last time when roads were used. Using CPLEX, the integer model proposed model is able to solve a real forest instance with 621 management units and 696 road segments to a 1% optimality gap. To deal with large scale tactical forest problems integrating harvest scheduling and road maintenance decisions, Bellavenutte et al. (2020) developed an optimization procedure that spatially partition the forest into clusters of stands, being the resulting subproblems solved by the solver Gurobi. The solution for the entire problem is obtained by an

iterative scheme that exchanges volume production targets between subproblems, in each iteration. They applied the proposed solution approach to a Brazilian forest of 125974 ha with 2935 stands.

To reduce the computational complexity of the integrated harvesting and forest road planning problem these problems can be addressed in sequence. First, a harvesting solution is obtained and then the underlying forest road network is built. The segregation of harvest scheduling of the road design can lead to suboptimal solutions, as the problems are clearly interdependent (Weintraub and Navon 1976, Guignard et. al 1998). Nevertheless, planning the road network and the timber transportation routes taking into account harvest scheduling is in itself a real challenge. In order to minimize road construction costs some authors proposed heuristics that use shortest path, minimum spanning tree, and/or Steiner tree-based algorithms (e.g., Weintraub et al. 1999, Stückelberger 2007). Epstein et al. (2006) and Legues et al. (2007) addressed the problem of selecting machinery location points, and of determining which forest segment road to build to connect these locations to forest exit roads. Specifically, Epstein et al. (2006) considered a single period un-capacitated network design model and proposed a heuristic approach which was applied to forest firms from Chile and Colombia. Legues et al. (2007) formulated this problem as a combination of a plant location problem and a fixed charge network flow problem. The authors developed a tabu search solution approach that embedded a minimum spanning tree procedure to determine the machinery location and the access road network. The resulting solution was further improved with a Steiner tree based procedure. Henningsson et al. (2007) addressed forest road upgrading needs and the concern with the accessibility to harvest areas under different weather conditions, by modeling the underlying problem as an un-capacitated fixed charge network flow problem including multiple assortments, several time periods and a set of different road classes. Aiming to minimize both transportation and road upgrading costs over a 10-years planning horizon they presented two MILP models. One model was based on arc flows and the other on path flows. Models were tested in a case study from a Swedish forest company considering a planning horizon of 10 years divided into two periods of five years. The MILP model based on path flows was further included in the optimization module of the decision support system RoadOpt, developed by Karlsson et al. (2006) as an extension of GIS ArcMap from ESRI to support forest road upgrading in Sweden. Meignan et al. (2012) addressed a road building problem aiming at minimizing road construction costs as well as harvesting costs. The authors pointed out the conflicting nature of these two objectives. On the one hand, a dense road network has high construction costs and yet facilitates the access to harvest areas resulting in lower harvesting costs. On the other hand, a sparse and cheaper road network makes it more difficult to access stands to be harvested thus increasing harvesting costs. The authors modeled the problem as the determination of a P-forest covering a subset of the graph vertices representing existing roads, exit roads and harvesting cells over a single periods and solved it with a Grasp metaheuristic. Najafi and Richards (2013) presented a MIP model to design a forest access system consisting of roads for trucking and access spurs for skidding while Bont et al (2015) considered a timber extraction

problem in a steep forest, with three different transportation modes: road, cable-yarding and helicopter. They proposed a MIP model with variables to decide the type of access and the type of harvesting technique.

This paper builds from this experience of model building and model solving to innovate the use of combinatorial optimization approaches to address forest road building and transportation. Specifically, the main contributions of this paper are: (1) a mixed integer linear programming (MILP) model to minimize the sum of road building, road maintenance, timber loading and transportation costs; (2) a Graph building module that receives as input information given by a Geographic Information System (GIS) software and defines a graph that characterizes the potential forest road network and the set of potential timber loading sites; (3) a Model solver module that iterates between a Estimation procedure aiming to tighten the big-M constraints of the MILP model and a Branch-and-bound Algorithm; (4) an agglomerative clustering algorithm that identifies groups of management units which are expected to share forest roads for timber transportation and provides information about upper bounds on timber flow capacities to the Estimation process. For testing purposes, we consider a large-scale forest management problem in Northwest Portugal involving 1137 stands.

2 Material and methods

2.1 Material

The forested landscape extends over 7623 ha located in Northwest Portugal, about 40 km southeast from Oporto (Fig. 1). It is a joint collaborative management area (ZIF)—ZIF Paiva—that has been classified into 1137 homogeneous management units (**MU**) (i.e., stands). The predominant species are eucalypt (*Eucalyptus globulus* Labill) and maritime pine (*Pinus pinaster* Ait.). Timber provides most forest management revenues in the area (Borges et al. 2017).

A solution setting a harvest scheduling plan for each management unit was obtained from solving a multi objective linear programming model which selects prescriptions for the management units. That solution is a point in a Pareto Frontier when optimizing 5 criteria and was selected by the stakeholders and the manager entity (Borges et al 2017). Its strategic temporal horizon encompassed ten 10-years periods. The plan conveys information about the stands to harvest in each period as well as about the volume to harvest in each stand. This plan conveys thus further the framework for tactical and operational decisions to provide access to the MUs being harvested so that timber may be transported from the loading sites in MUs harvested to the national and municipal roads and afterwards to sawmills and pulp factories. Specifically, road building, road maintenance and timber loading location decisions in any given period impact the decision space in subsequent periods. This impact is better captured by a model that encompasses all ten 10-years periods of the strategic horizon.

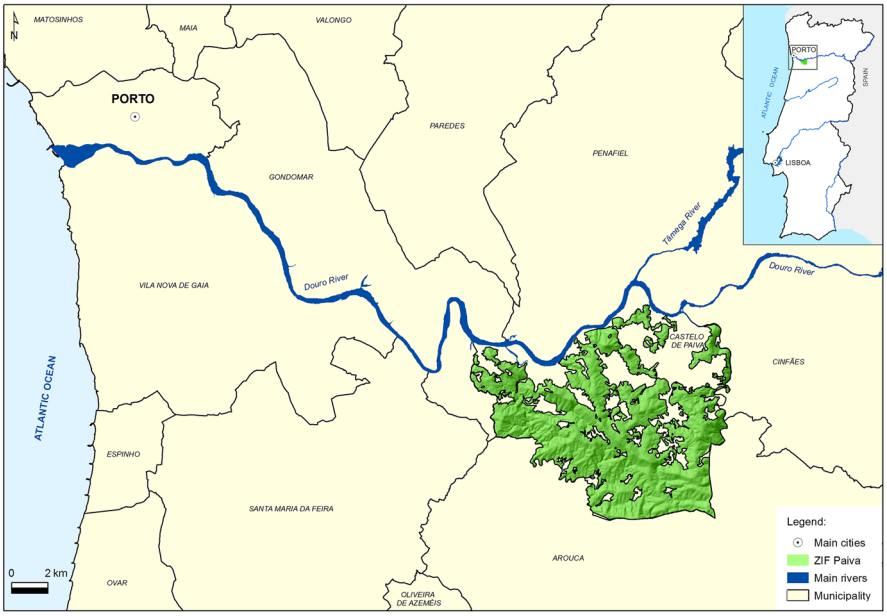

Fig. 1 Location of the ZIF Paiva case study area in Portugal

2.1.1 Network design

In each planning period, one needs to build or maintain forest roads that link stands to be harvested to pre-existing roads under the supervision of municipalities, so that timber may be transported to sawmills and pulp plants. Pre-existing roads (national, municipal and rural) crossing and leaving the forest under management (Fig. 2) will be denoted as **exit roads**, in contrast to the roads that are to be built or maintained, which will be denoted as **forest roads** (or **potential forest roads**). Some stands in the forested landscape are bordered by parcels of land that do not belong to ZIF Paiva. However, it may be necessary to cross the boundary of these parcels to reach exit roads. Consequently, they will be included in the current study and denoted as **non-managed land units (NMU)**.

Timber resulting from harvesting operations is forwarded to the boundary of the stands so that it may be loaded on trucks. Available loading locations will be called **timber loading sites**. Forest roads linking timber loading sites to exit roads will be built at the boundary of MUs or NMUs.

Geographical and topographical information characterizing MUs, NMUs and exit roads was processed by GIS (Geographic Information System) routines, using ArcGIS 10.6, to identify the potential forest road network as well as the timber loading sites. The data used for the analysis encompassed the MU and NMU (polygons) and the existing road network (polylines). Since forest roads will be built at the boundaries of the MUs or NMUs, the polygon corresponding to each MU (or NMU) was covered by a set of vertices, numbered sequentially, and by a set of edges linking

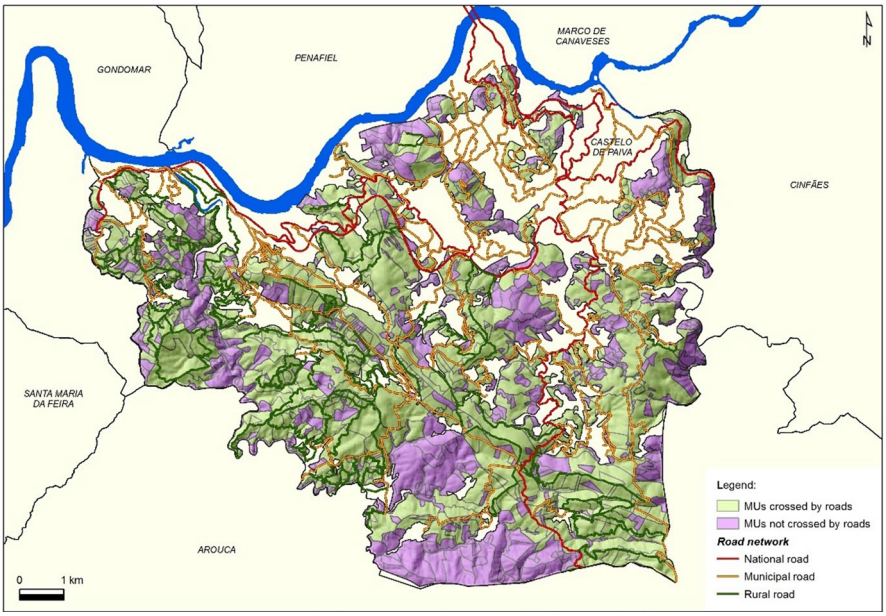

Fig. 2 ZIF Paiva road network and management units

pairs of sequentially numbered vertices, as well as the initial and the last vertex. Polylines corresponding to the pre-existing road network were also covered by vertices and edges linking adjacent vertices. Points in the intersection of the pre-existing road network with the boundaries of MUs, or NMUs, represent forest exit points for the trucks that travel along the boundaries of the MUs and NMUs. Thus, they were identified and denoted as exit road vertices (Fig. 3).

To identify potential timber loading sites and to estimate road construction and maintenance costs as well as timber transportation costs up to the exit roads each vertex was associated with information, identifying the MUs (NMUs) it belongs, its geographical coordinates as well as topographical features such as slope and altitude (Fig. 4).

2.1.2 Cost parameters

Costs were estimated and assigned to vertices and to edges. All costs take into account the slope of the terrain as well as prices provided by the Portuguese Forest and nature Conservation Institute—ICNF (2018) (Table 1. Costs according to slope). We considered three classes of slope: low (< 5%), medium (5–25%) and high (> 25%) (Table 1. Costs according to slope).

In each management unit being harvested in a specific period, timber loading will occur in a vertex of its boundary. Consequently, loading costs were assigned to each vertex in the boundary of a MU. Loading costs are influenced by slope, since an increase in slope leads to a decrease in productivity (Ackerman et al. 2018). The practice in the region is to subcontract the timber loading operation, and in general,

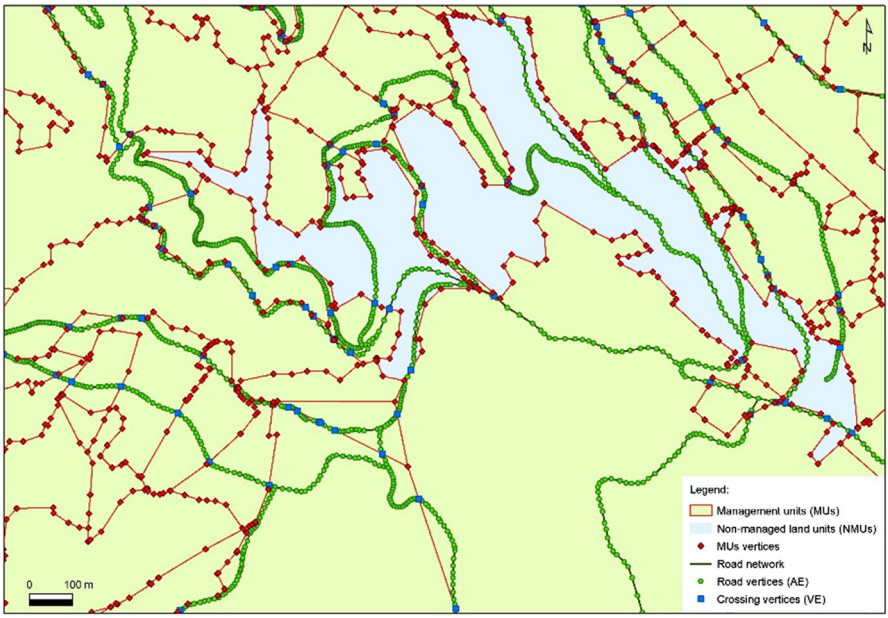

Fig. 3 Vertices and edges covering MUs boundaries and the road network in a sub area in ZIF Paiva

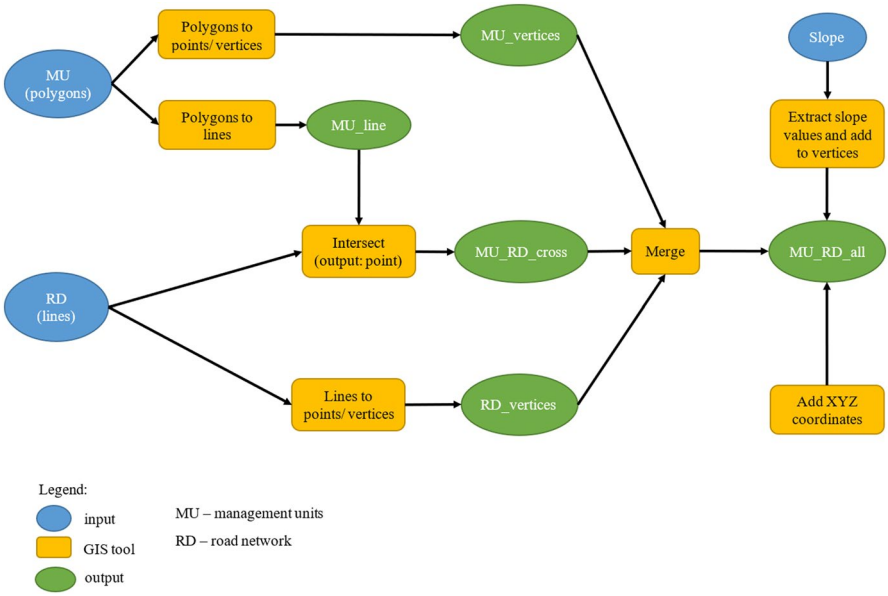

Fig. 4 Flow chart for the characterization of management units and road network vertices and identification of potential roads and timber loading sites

Table 1 Costs according to slope

Slope classes (%)	Loading (€/t)	Building (€/km)	Maintenance (€/km)
[0–5]	1.30	1850.40	925.20
[5–25]	1.95	4163.40	1624.10
> 25	2.60	6476.40	2313.00

Source: *adapted from ICNF (2018)*

the fees charged depend on the slope and follow the cost benchmark published by ICNF (2018). We considered three loading cost values, one for each slope class. The lowest load cost (1.30 €/t) was associated with the lowest slope class and the highest cost (2.60 €/t) with the highest slope class. The medium slope class (5–25%) was associated with an average loading cost (i.e. 1.95 €/t).

Road building, road maintenance and transportation costs were assigned to each edge {i,j}. These costs take into account the Euclidean distance between vertices i and j, the slope of the terrain and the costs published by ICNF (2018). Indeed, vertices i and j are vertices of the polygon that represents a MU. These polygons, obtained via ArcGIS, accurately describe the MU polygon in the terrain. Consequently, the Euclidean distance gives a good estimate of the real distance between i and j.

The maximum and minimum road building and maintenance costs provided by ICNF (2018) were associated to the lowest and the highest slope classes. Moreover, an average value, between the minimum and maximum costs, was associated to the medium slope class (5–25%) vertices (Table 1. Costs according to slope). ArcGIS includes the Slope tool that identifies the slope for each cell by calculating the maximum rate of change in value from that cell to its neighbors. Thus, preliminary building and maintenance costs were associated to each vertex according to the corresponding slope. Then, for each pair of adjacent vertices, {i,j}, we computed the average value of the corresponding building and maintenance costs, respectively denoted by, Avbij and Avmij. The building (maintenance) cost assigned to the edge {i,j} is obtained by multiplying Avbij (Avmij) by the distance between vertices i and j.

ICNF (2018) only provides information on transportation costs per unit of weight. For a truck with capacity of 25 ton, ICNF (2018) considers a transportation cost of 5.80 (€/t) or 19.20 (€/t) depending on whether the distance to the industry is less than 30 km or greater than 100 km. These values give a cost of 145€ and 480€ for a truck of 25 ton moving 30 km and 100 km, respectively, and suggested a minimum and maximum values of 4.80 (€/km) and 4.83 (€/km). The minimum (maximum) value was assigned to vertices with the lowest (highest) slope and an average value was assigned to average slope vertices. Transportation costs associated to edge {i,j} were calculated taking into account these preliminary costs assigned to vertices i and j, the distance between these two vertices and an average distance to the mill. The latter was estimated by the geographical information system (ArcGIS) taking into account the distance between each vertex and the mill.

We considered a discount rate of 3% and costs taking place in the midpoint of the planning period. Consequently, costs in period t are given by the expression $C^t = 1.03^{5+10t}C_0$, where C_0 is the cost in the beginning of period 1.

A graph representing the potential forest road network is created. Each MU boundary vertex is characterized by a loading cost and each road edge is characterized by a building, a maintenance and a transportation cost. This graph is the basis upon which a mathematical model is developed to decide which roads to build/maintain as well as to define the timber flow in each segment road in each period.

2.2 Methods

In each time period, decisions must be made on building new forest roads or maintaining existing roads, as well as setting timber load locations and defining timber routing. Although these decisions in one period may be considered at tactical level, they are interconnected with the ones made in subsequent periods. Hence, the mathematical model should cover the whole strategic time horizon.

We propose a MILP model to address the road building, road maintenance, timber loading location sites and timber transportation problem (RoadFor, for short). It includes big-M constraints to relate road building and maintenance with transportation decision variables. The use of big-M constraints leads to computational drawbacks, while using commercial solvers, related to numerical instability and weak linear programming bounds. Thus, in order to tighten the big-M constraints, we develop a heuristic approach that estimates, in each period, the maximum flow of timber transportation over the edges.

2.2.1 Mathematical model

Let U be a set of management units and FMP be a forest management plan that schedules harvests in these units over a temporal horizon that includes $|T|$ periods ($T = \{1, \ldots, |T|\}$). We denote by TIM_u^t the amount of timber (ton) to harvest in period t in management unit u. The total amount of timber to harvest in period t in the forest is given by $\mathrm{TIM}B^t = \sum_{u \in U} TIM_u^t$.

Roads construction and maintenance will be dealt on the undirected graph G = (V,E). The set E includes the edges defining MUs and NMUs boundaries as well as the edges connecting exit road vertices to some MU or NMU vertices. The set of vertices V is the union of three subsets, $\mathrm{V} = \mathrm{VP} \cup \mathrm{VQ} \cup \mathrm{VE}$. VP is the subset of vertices that cover boundaries of management units in U. Let $\mathrm{VP}(u) \subseteq \mathrm{VP}$ denote the set of all vertices in the boundary of management unit u. VE consists of exit road vertices. An exit road vertex may belong to the boundary of a MU or may be located outside the boundary of any MU. Set VQ includes intermediate vertices that may link management units to exit roads along the boundary of NMUs. Note that, $\mathrm{VP} \cap \mathrm{VQ} = \emptyset$ and that we consider $\mathrm{VE} \cap \mathrm{VQ} = \emptyset$. An edge $\{i, j\} \in E$ represents a potential road segment linking vertices $i, j \in \mathrm{V}$, which can be built or maintained in period t. Figure 5 provides a small example to illustrate how the graph G = (V,E) is built.

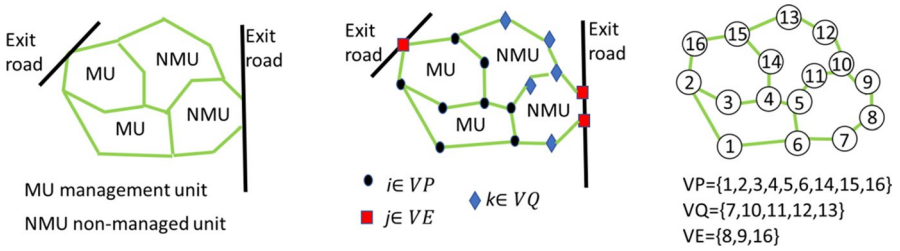

Fig. 5 From MUs, NMUs and exit roads to a graph G = (V,E), with V = VP U VQ U VE

In order to address timber transportation, we need to know the flow direction in each road segment road. To denote a directed edge we use the term arc. Moreover, while an edge linking nodes i and j is represented by {i,j}, an arc directed from i to j is represented by (i,j). We consider a set of arcs AV, such that (i,j) ∈ AV if timber can flow from i to j in the potential road segment {i,j}. Flow is in general possible in both directions. However, the flow from an exit road vertex to any other vertex does not need to be considered, since whenever a truck reaches an exit road vertex it will follow the main roads.

To easily handle the choice of loading sites, we consider a set of additional vertices, VU, where each vertex represents a management unit. The resulting vertex set is denoted by N = V ∪ VU. Associated to each vertex $u \in$ VU there is a set of arcs linking u to the vertices of the boundary of the respective management unit (see Fig. 6).

To address the timber loading sites and timber routing optimization subproblems we consider the digraph H = (N,A) where A = AU ∪ AV, AU = $\{(u,i) : u \in VU, i \in VP(u)\}$ and AV = $\{(i,j) : i \in V \backslash VE, j \in V\}$.

For each period $t \in T$, the following (discounted) costs are known: CB^t_{ij} is the cost of building road segment corresponding to edge {i,j}; CM^t_{ij} is the cost of maintaining road segment corresponding to edge {i,j}; CT^t_{ij} is the transportation cost of one ton of timber in road segment corresponding to arc {i,j}; CL^t_i is the cost of loading one ton of timber in vertex i.

We developed a mixed integer linear programming model aiming to minimize road building and maintenance costs, timber loading and transportation costs over the planning horizon T.

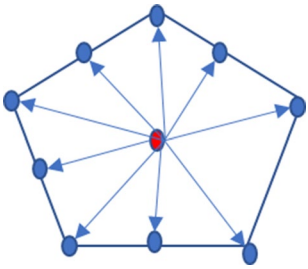

Fig. 6 Arcs linking vertex u to the boundary vertices and edges representing the boundary of management unit u

Three sets of decision variables are considered:

$$w_{ij}^t = \begin{cases} 1, \ \textit{if road } \{i,j\} \text{ is built in period } t \\ 0, \ \text{otherwise} \end{cases}, \{i,j\} \in E, \ \mathrm{t} \in T;$$

$$y_{ij}^t = \begin{cases} 1, \ \textit{if road } \{i,j\} \text{ is maintained in period } t \\ 0, \ \text{otherwise} \end{cases}, \{i,j\} \in E, \ \mathrm{t} \in T;$$

$$f_{ij}^t = \text{timber flow } (\textit{ton}) \text{ in arc } (i,j) \text{ in period } t, (i,j) \in A, \mathrm{t} \in T.$$

Note that, f_{ui}^t gives the amount of timber harvested in management unit u, in period t, loaded from node i.

$$\min \sum_{t \in T} \sum_{\{i,j\} \in E} CB_{ij}^t w_{ij}^t + \sum_{t \in T} \sum_{\{i,j\} \in E} CM_{ij}^t y_{ij}^t + \sum_{t \in T} \sum_{(u,i) \in AU} CL_i^t f_{ui}^t + \sum_{t \in T} \sum_{(i,j) \in AV} CT_{ij}^t f_{ij}^t \tag{1}$$

Subject to:

$$\sum_{i:(u,i) \in AU} f_{ui}^t = TIM_u^t, u \in \mathrm{VU}, t \in T \tag{2}$$

$$\sum_{u:(u,i) \in AU} f_{ui}^t + \sum_{j:(j,i) \in AV} f_{ji}^t = \sum_{j:(i,j) \in AV} f_{ij}^t, \quad i \in V \backslash VE, t \in T \tag{3}$$

$$f_{ij}^t + f_{ji}^t \leq M_{ij}^t \times \left(w_{ij}^t + y_{ij}^t \right), \quad \{\mathrm{i,j}\} \in E, i,j \in V \backslash VE, t \in T \tag{4}$$

$$f_{ij}^t \leq M_{ij}^t \times \left(w_{ij}^t + y_{ij}^t \right), \qquad \{i,j\} \in E, j \in VE, t \in T \tag{5}$$

$$y_{ij}^t \leq w_{ij}^{t-1} + y_{ij}^{t-1}, \qquad \{i,j\} \in E, t \in T \tag{6}$$

$$y_{ij}^0 = 0, \qquad \{i,j\} \in E \tag{7}$$

$$w_{ij}^0 = 1, \textit{if road} \{i,j\} \textit{exists in the beginning of the horizon}, w_{ij}^0 = 0, \textit{otherwise} \tag{8}$$

$$f_{ij}^t \geq 0, \qquad (i,j) \in A, t \in T \tag{9}$$

$$w_{ij}^t, y_{ij}^t \in \{0,1\}, \qquad \{i,j\} \in E, t \in T \tag{10}$$

The objective function (1) incorporates road building and maintenance costs, timber loading and transportation costs along the planning horizon T.

Constraints (2) ensure the timber produced in each management unit in each period is loaded in one of the vertices belonging to the unit. Constraint (3) are flow conservation constraints, stating that, for each non exit vertex i and each time period t, the amounts of timber entering and leaving i are equal.

Constraints (4) and (5) force a road to be built or maintained if timber flows through it. M_{ij}^t is an upper bound on the maximum amount of timber that can flow through road {i,j} in period t. This value can be set to $\text{TIM}B^t = \sum_{u \in U} TIM_u^t$ for every road. Therefore (4) and (5) are not capacity constraints, as all timber produced in the forest in any period can flow through road {i,j} in period t.

Constraints (6) ensure that only existing roads may be considered for maintenance in any period. Constraints (7) and (8) establish the state of the roads in the first period.

In this model there is no need to enforce a single loading site in each management unit. Indeed, since there is no capacity on the amount of timber that can flow in any arc there exists an optimal solution of the model such that for each unit and time period, timber loading is performed in only one site (see e.g. Magnanti and Wolsey, 1995).

2.2.2 Solution approach

To solve the RoadFor problem we propose a **Heuristic algorithm** that relies on the MILP model (1) to (10). This mathematical model includes a set of big-M constraint parameters, (4) and (5). Large values of M_{ij}^t, $t \in T$ lead to weak linear programming bounds and may cause numerical instability in model solving. This instability may lead to huge computational costs (e.g., CPU time) to reach optimality or, even, to reach feasible solutions. The Heuristic algorithm encompasses two modules: Graph building and Model solver. The Model solver module includes a big-M Estimation procedure and a Branch and Bound solution method.

2.2.2.1 Graph building The Heuristic algorithm starts with a **SpatialNetworkToGraph procedure** that transforms the information given by the GIS into a graph characterizing potential forest roads and exit roads by a set of vertices and a set of edges. The GIS software covers the polygon of each MU, or NMU, with a sequence set of vertices. The edges are then established according to this sequence. Consequently, in the information provided by the GIS software, each vertex and each edge belong to a single MU, or NMU, boundary. However adjacent units share vertices and edges. Boundaries shared by two, or more, management units have different "named" vertices with the same spatial coordinates, or pairs of vertices whose distance between them may be considered negligible (see Fig. 7).

The **SpatialNetworkToGraph procedure** removes duplicated vertices and re-adjusts edges linking adjacent vertices. Non duplicated vertices are then re-assigned to the different management units they cover.

Moreover, we need to integrate the GIS polygons (e.g., MU and NMU) and GIS lines corresponding to pre-existing roads (national, municipal and rural). For each MU (NMU) crossed by an existing road we identify the exit road vertices that are

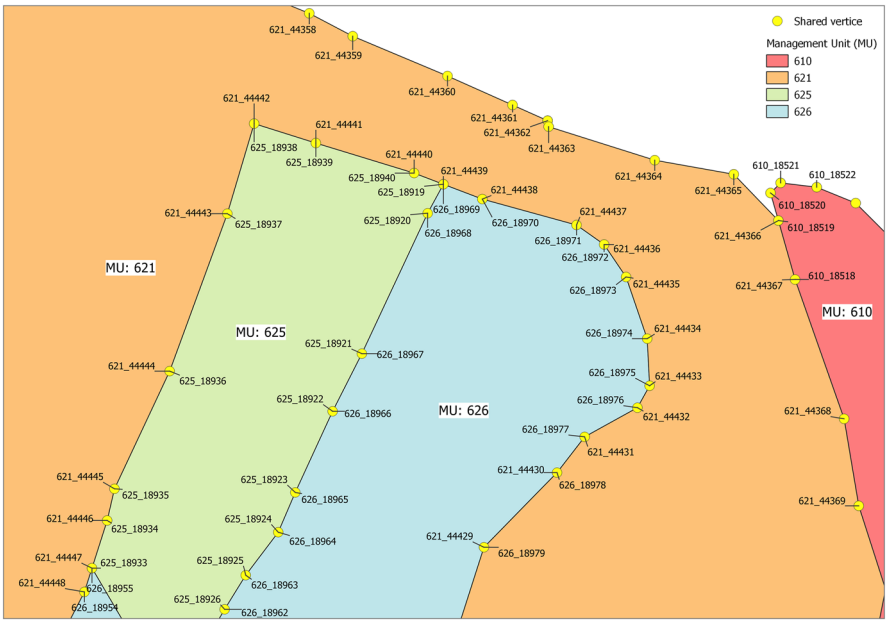

Fig. 7 Example of shared vertices and edges of the adjacent units 610, 621, 625 and 626

in the intersection between the road and the boundary of the MU (NMU). Each exit road vertex is assigned to the MUs or NMUs it intersects. For each exit road vertex *e* assigned to a MU, *u*, the SpatialNetworkToGraph procedure determines the edge {i,j} with i, j ∈ VP(*u*) for which the perpendicular distance from *e* to {i,j} is minimum. Two new edges, {i,e} and {e,j}, are created and edge {i,j} may be removed. The same procedure is applied to link exit road vertices to NMUs, whenever adequate, to complete the definition of graph G = (V,E).

The **GraphSimplification procedure** simplifies graph G = (V,E) while maintaining its overall connectivity. It works in two steps. In step 1, the procedure reduces the cardinality of vertex set V and the edge set E (arc set A) by contracting subsets of vertices with degree two. Consider vertex i such that degree(i) = 2. Vertex i has two adjacent vertices further denoted by k and j (Fig. 8). There is a single path from k to j passing through i, which makes vertex i redundant for timber transportation and, whenever the vertex is not suitable for timber loading, it can be removed. We

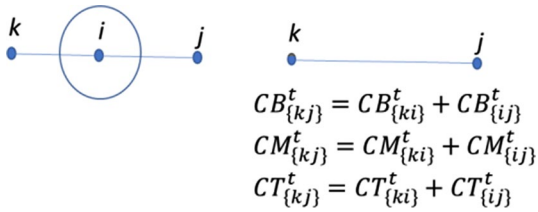

Fig. 8 Removing a degree-2 vertex and merging the incident edges

remove vertex i and merge the two incident edges, {k,i} and {i,j}, into a new edge linking {k,j} (Fig. 8). Building, maintenance and transportation costs assigned to {k,j} are the sum of the corresponding two original edges costs.

However, vertex i (degree(i) = 2) may represent a promising timber loading site and its removal may lead to a more expensive solution. Thus, the GraphSimplification procedure implements this removal using an approach that relies on the presence or not of exit road vertices in the boundary of the MU. In the case of management units that include exit road vertices in their boundary timber loading is expected to take place in one of the latter (or very close to) to avoid additional building, maintenance and transportation costs. This feature suggests the partition of the vertex set VU into two subsets UVE (management units with vertices from VE) and UWVE (management units without VE vertices). Vertex $u \in UWVE$ iff $\mathrm{VP}(u) \cap \mathrm{VE} = \emptyset$, otherwise it belongs to UVE.

For a management unit $u \in UVE$, the GraphSimplification procedure removes all vertices $\mathrm{i} \in V\mathrm{P}(u) \backslash \mathrm{VE}$ with degree(i) = 2, since timber loading will occur in an exit road vertex. Note that, if the management unit has only two vertices k, j with degree greater than two then the removal can end with multiple edges linking vertices k and j. In such case, if two edges share the same transportation, maintenance and building costs, then the procedure deletes one of them. Otherwise, it keeps the one corresponding to the minimum sum of transportation, maintenance and building costs.

In the case of a management unit $\mathrm{u} \in U\mathrm{WVE}$ the removal of a vertex i with degree two is subject to an additional condition requiring that timber loading cost(i) > L, where L is a pre-defined parameter representing a large timber loading cost. Vertex with large loading costs assigned are not expected to be chosen for timber loading in an optimal solution. So, the Graph Simplification procedure removes all vertices $\mathrm{i} \in V\mathrm{P}(u) \backslash \mathrm{VE}$ such that (degree (i) = 2 and loading cost(i) > L).

Vertices from VQ are all intermediate nodes for timber transportation. Thus, the procedure removes all vertices $\mathrm{i} \in V\mathrm{Q}$, when degree(i) = 2.

The SpatialNetworkToGraph procedure links an exit road vertex to a MUs or a NMUs boundary at the expense of additional edges. In step 2, the GraphSimplification procedure uses two criteria based on arc distances to eliminate some arcs.

Consider vertices $i \in (VP \cup VQ) \backslash VE$. Suppose there are edges $\{i, j_1\}, \{i, j_2\}, \ldots, \{i, j_k\}$ where $j_1, j_2, \ldots, j_k \in VE$ and $dist(i, j_1) \leq dist(i, j_2) \leq \cdots \leq dist(i, j_k)$. If, the shortest path distance is used to reach an exit road from i then all edges $\{i, j_2\}, \ldots, \{i, j_k\}$ may be removed (left side of Fig. 9).

Additionally, consider the situation where two adjacent vertices i_1, i_2, with equal loading costs, $CL(i_1) = CL(i_2)$, are adjacent to the same exit road (right side of

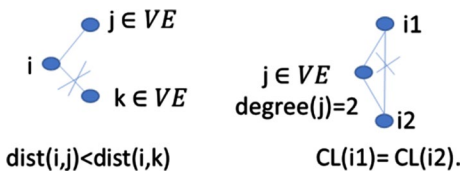

Fig. 9 Removing some edges based on distance criteria

Fig. 9). If $CL(i_1) = CL(i_2)$ and $\text{dist}(i_1, i_2) > \max\{\text{dist}(i_1, j), \text{dist}(i_2, j)\}$, then the edge (i_1, i_2) is not expected to be crossed in a path to reach the exit road vertex j.

Vertices without incident edges, are removed from the vertex set V.

The criteria described above are based on edges distances. However, each edge has three types of costs associated to it (e.g., building, maintenance and transportation costs). In order to ensure that we are not neglecting a significant number of promising solutions we considered the set of all edges and computed the correlation coefficient between the distance and building, maintenance, transportation costs having obtained the values 0.972, 0.981, 0.976, respectively.

2.2.2.2 Model solver To solve the resulting RoadFor problem we propose a Model Solver Module which iterates between a Estimation Procedure and a Branch&Bound algorithm. The **Estimation procedure** determines estimates for M_{ij}^t, in each period t, aiming to tighten the big-M constraints (4) and (5) while ensuring the feasibility of (near) optimal solutions of the resulting MILP model, which is then solved by a Branch&Bound algorithm.

Big-M constraints link timber flow continuous decision variables with binary road building and maintenance decision variables ensuring that timber is transported through arcs that have been built or maintained. Thus, the value of M_{ij}^t should be large enough to ensure that all timber harvested in period t may be transported through the road arc (i,j). An eligible value to an estimate of M_{ij}^t is given by the total timber harvested in all management units, in period t. That is, no feasible solution is discarded if one considers, for each period t, $M_{ij}^t = \text{TIMB}^t = \sum_{u \in U} TIM_u^t, \forall (i,j)$. However, if TIMB^t is a large value then the resulting MILP problem may remain difficult to solve due to weak lower bounds in the Branch&Bound algorithm. To overcome this drawback, we propose a procedure that takes advantages of specific features of the problem to tighten the value of TIMB^t without cutting off promising solutions.

On the one hand, due to road building and maintenance costs, we anticipate that MUs belonging to UVE which are crossed by municipal roads only use forest roads if these roads have already been built or maintained to service other MUs. In fact, MUs belonging to UVE include in their boundary exit road vertices near which timber loading can take place avoiding extra costs on building, maintenance and timber transportation. Consequently, we will exclude MUs belonging to UVE from the Big-M estimation process. On the other hand, MUs $\in$ UWVE with harvesting activities in a period t are usually spread along the forest and have several alternative exit road vertices, also scattered along the forest, to exit the forested landscape. It is not expected that all timber harvested in period t flows through the same forest road arc. These issues suggest the development of a Estimation procedure based on the identification of clusters of MUs belonging to UWVE such that, MUs in different clusters are not expected to share the same forest roads for timber transportation.

To initialize the set of clusters, the Estimation procedure considers groups of adjacent MUs belonging to UWVE (Fig. 10). To this end, it builds graph $G_a = (UWVE, E_a)$. The vertex set of G_a includes vertices $u \in UWVE$ representing management units without exit road vertices in their boundary. There is an edge $e = \{u_i, u_j\}$, between vertices

Fig. 10 A connected component of G_a

u_i and u_j, iff $\mathrm{VP}(u_i) \cap \mathrm{VP}(u_j) \neq \emptyset$, that is, whenever management units u_i and u_j are adjacent. The Estimation procedure starts by identifying the set of connected components of G_a. Each connected component includes a subset of adjacent management units or eventually a single management unit, not crossed by exit roads.

Starting with each connected component as a cluster, we may devise a clustering algorithm that merges clusters based on the knowledge that in some routes to exit road vertices some edges may be shared by vehicles that transport timber from management units of different connect components (Fig. 11).

The proposed clustering algorithm associates with each cluster a set of attractive exit road vertices. Then it uses an agglomerative scheme to merge pairs of clusters based on the percentage of exit road vertices shared by two clusters. Given a final set of clusters, one can estimate the value of $M^t_{ij}, (i,j) \in E, t \in T$ by considering the set of MUs in each cluster.

The Estimation procedure may be described by the following steps.

Step 1—Initialization Let CP denote the set of all clusters, covering all MUs from UWVE, each cluster corresponding to a connected component in graph G_a. Let $\mathrm{C}P_h$ denote cluster h.

Step 2—Identification of attractive exit road vertices For each vertex $i \in \mathrm{VP}$, one determines the shortest path from i to each $v \in \mathrm{VE}$. Then, we pick the exit road vertices that are the sink of the k shortest distance paths. Let us save these exit road vertices in set $\mathrm{VE}_k(i)$. Let us denote by $\mathrm{VE}(CP_h)$, the set of all exit road vertices selected for all MUs in cluster $\mathrm{C}P_h$. That is, $\mathrm{VE}(CP_h) = \cup_{\mathrm{i} \in \mathrm{VP}(\mathrm{CP_h})} \mathrm{VE_k(i)}$. At the end of this step each cluster CP_h has associated set $\mathrm{VE}(CP_h)$ including the exit road vertices that are within a short distance of reach from MUs in cluster $\mathrm{C}P_h$.

Step 3—Building a merging list The clustering algorithm creates new clusters $\mathrm{C}P_{nm}$ by merging clusters $\mathrm{C}P_n$ and CP_m sharing at least a pre-defined percentage, denoted by *Perc*, of the attractive exit road vertices assigned in step 2. For each pair of clusters $\mathrm{C}P_n$ and CP_m a percentage, $per(m,n)$, is calculated as the quotient between the number of vertices in $(\mathrm{VE}(CP_n) \cap \mathrm{VE}(CP_m))$ and the number of vertices

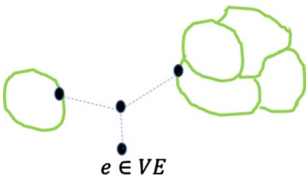

Fig. 11 A cluster including two connected components

in $\left(\mathrm{VE}\left(CP_n\right) \cup \mathrm{VE}\left(CP_m\right)\right)$. Note that, if clusters $\mathrm{C}P_n$ and CP_m do not share any exit road vertices, then $\mathrm{VE}\left(CP_n\right) \cap VE\left(CP_m\right) = \emptyset$ and $per(m,n) = 0\%$, while if $\mathrm{VE}\left(CP_n\right) = \mathrm{VE}\left(CP_m\right)$ then $per(m,n) = 100\%$. If $per(m,n) > Perc$, then we save the information $\left(CP_n, \mathrm{C}P_m, per(m,n)\right)$ into a merging list including the candidate clusters for merging. After all pairs of classes have been analyzed, if the merging list is not empty continue to step 4. Otherwise, no changes in the clusters of set CP occur and proceed to **step 5**.

Step 4—Merging clusters Following a top-down strategy, clusters $\mathrm{C}P_i$ and CP_j are merged into a new cluster CP_{ij} and, all the components including cluster $\mathrm{C}P_i$ or cluster CP_j are removed from the merging list. Moreover, set CP is updated by removing clusters $\mathrm{C}P_i$ and CP_j and including the new cluster CP_{ij}. Additionally, set $\mathrm{VE}\left(CP_{ij}\right)$ is also updated to $\mathrm{VE}\left(CP_{ij}\right) = \mathrm{VE}\left(CP_i\right) \cup \mathrm{VE}\left(CP_j\right)$. Return to Step 3.

Step 5—Estimation For each new cluster $\mathrm{C}P_i \in CP$, we compute $M_i^t = \sum_{u \in CP_i} TIM_u^t$. For each period $t \in T$ the estimate of $M_{ij}^t, (i,j) \in E, t \in T$ is given by $m^t = \max_i M_i^t$.

The values $m^t, t \in T$, determined in Step 5 are assigned to $M_{ij}^t, \forall (i,j) \in E, t \in T$ in constraints (4) and (5) of the mathematical model presented in Sect. 2.2.1, establishing an upper capacity on the total amount of timber that can flow through the arcs $(i,j) \in A, t \in T$.

The resulting MILP problem is then solved by a Branch&Bound procedure that includes two stopping criteria, a MIP gap tolerance and a time limit, to guarantee that a commercial solver ends with an optimal or near optimal solution.

The output of the Estimation procedure depends on the values assigned to parameters k, which controls the selection of attractive exit road vertices, and $Perc$, that determines the candidate clusters for merging. Changes in these parameter values may lead to different final sets CP and to different estimates. The Model solver we propose follows an iterative scheme by considering different values for these parameters. We consider that parameter k can take values in set $\mathrm{S} = \{1,2,\ldots,\mathrm{K}\}$ and parameter $Perc$ can take values in set $\mathrm{P} = \{p_1, p_2, \ldots, p_q\}$, $0 \le p_j \le 1, j = 1, \ldots, q$ and we define a subset $SP \subseteq S \times P$ containing the pairs of values that k and Perc will assume in each iteration. We initialize the upper bound on the optimal value of the objective function (1) to $\mathrm{UB} = \infty$. In iteration r, the Model solver considers the pair of values $\left(s_r, p_r\right) \in SP$ and executes Steps 2 to 5. Then, with the resulting set of estimates at handit runs the Branch&Bound procedure. If the value of the resulting solution is less than the current upper bound then it updates the value of UB and saves the corresponding solution.

The Heuristic Algorithm receives as input a harvest scheduling plan covering a time horizon of several periods, GIS data characterizing the forest landscape under study and information on forest road building and maintenance costs as well as timber loading and transportation costs. This information is processed by the Graph building module giving as output a graph describing the RoadFor problem. The subsequent Model solver module receives as input the graph obtained in the previous step, the set SP including the pairs of parameters (k, $Perc$) to be

considered during the Estimation procedure and the mathematical model proposed in Sect. 2.2.1. The heuristic Algorithm gives as output a RoadFor (near) optimal solution containing information on which forest roads to build and/or maintain, timber loading sites and timber transportation routes for MUs with harvest activities, in each period. The main steps of the Heuristic algorithm can be summarized in Fig. 12.

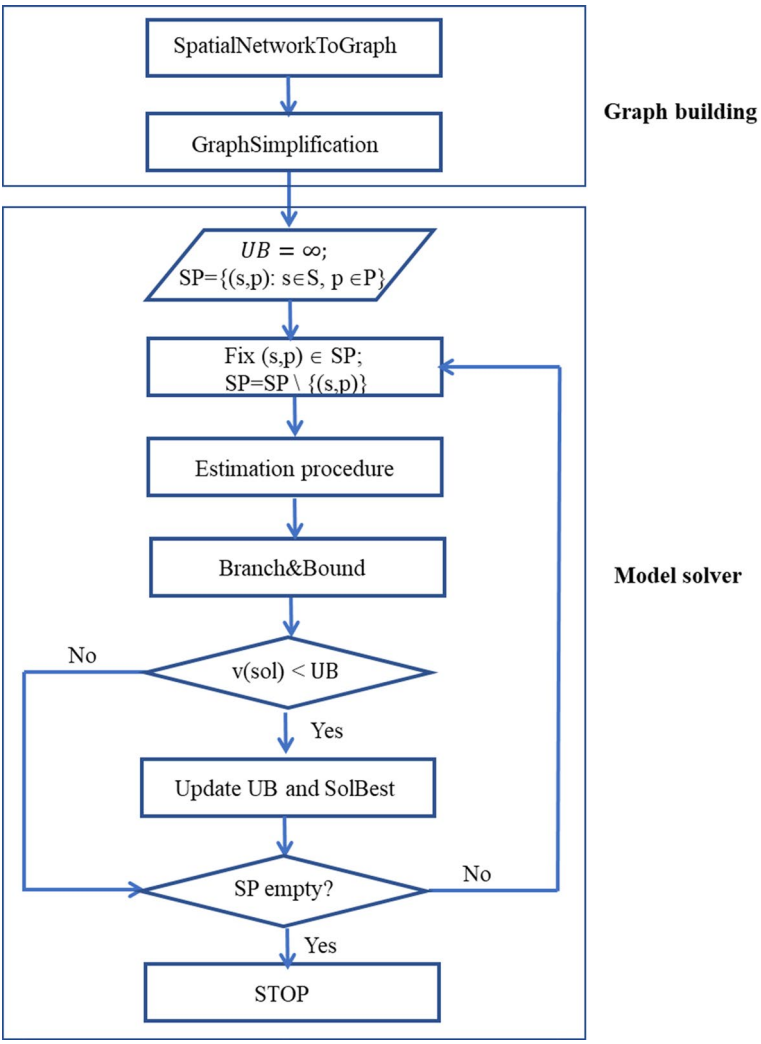

Fig. 12 Heuristic algorithm

3 Computational results and discussion

After applying the Estimation Procedure, to solve the resulting MILP models in the Branch&Bound Step we used IBM ILOG CPLEX Optimization Studio V12.8.0 and ran on an Intel R Core TM i7-2620M CPU 2.70 gigahertz RAM 8.00 gigahertz. We considered the default options for the branch-and-bound algorithm except the mixed integer optimality gap tolerance which was set to 0.03 and the time limit which was set to 72 h.

3.1 Computational results

Table 2 displays information on the number of vertices, edges and arcs along the steps of the Heuristic algorithm. The row denoted by "GIS" gives the number of vertices, edges and arcs provided by the GIS software to cover the boundary of MUs, NMUs. It also shows the number of exit road vertices that belong to the boundary of MUs or NMUs but do not consider edges and arcs incident into the exit road vertices. The SpatialNetworkToGraph procedure removes duplicated information and links exit road vertices to MUs and NMUs vertices (row "SpatialNetworkToGraph"). All the above information is updated further by the GraphSimplification procedure to build graph $G = (V, E)$ and digraph $H = (N, A)$. The four columns under the identification "#vertices" show the number of vertices in the additional vertex set VU as well as in each subset of the vertex set $V = VP \cup VQ \cup VE$. Column "#edges" gives the number of edges available in each step of the heuristic algorithm. Finally, columns "AU" and "AV" describe the arc set $A = AU \cup AV$, in terms of the number of arcs in each of its subsets.

The harvest scheduling plan provides values to estimate M^t_{ij} (further denoted by M^t, since we do not distinguish by edge) based on the sum of timber tons harvested in each period, $\mathrm{TIMB}^t = \sum_{u \in U} TIM^t_u$ and based on the sum of timber tons harvested in management units without exit road vertices in their boundaries in each period, $TIM^t_{UWVE} = \sum_{u \in UWVE} TIM^t_u$ (Table 3).

The Estimation procedure considers two parameters, k and *Perc*. Parameter k establishes the number of attractive exit road vertices that should be associated to each vertex of the MUs belonging to *UWVE*. For example, if k = 1 then, on associates to each vertex in the boundary of the MU, the shortest distance exit road vertex.

Table 2 The number of vertices, edges and arcs along different steps of the heuristic algorithm

	#vertices				# edges	# arcs	
	VU	VP	VQ	VE		AU	AV
GIS	–	46,576	3943	2949	50,519*	–	101,038*
Spatialnetworktograph	–	27,455	2429	1794	33,635	–	63,228
Graphsimplification	1137	10,785	275	1653	13,086	19,018	23,079

Values with * do not include edges/arcs incident to exit road vertices

Table 3 Amount of timber harvested in each period, considering all management units ($TIMB^t$) and considering managements units without exit road vertices (TIM^t_{UWVE})

Period	1	2	3	4	5	6	7	8	9	10
$TIMB^t$	112,680.1	121,250.9	233,409.2	189,455.6	167,444.3	151,406.9	154,775.4	164,316.6	164,316.6	64,716.8
TIM^t_{UWVE}	28,534.7	55,615.6	92,233.5	76,018.6	62,510.6	63,885.4	62,049.6	61,818.7	90,959.6	27,073.9

To obtain a diverse set of estimates of M^t, we have considered three different values for k, k = 1,2,3.

Parameter *Perc* defines the minimum percentage of k shortest distance exit road vertices shared by two connected components in order to be merged into the same class. We have considered $Perc = 0.01, Perc = \overline{per}$ ($\overline{per}$ is the average percentage of exit road vertices shared by two clusters in Step 1 and Step 2 of the Estimation procedure) and $Perc = 0.25$.

The final set of clusters may depend on the sequence of pairs of clusters submitted to the merge in Step 4 of the Estimation procedure. Consequently, for each value assigned to k and *Perc,* we have re-arranged the merging list according to two different criteria. The components of the merging list are sorted by descending order of *per*. In alternative, the components are randomly chosen for the merging process. In summary, 18 iterations of the Estimation procedure were executed. Note that, the same set of estimates may be obtained in different iterations. The resulting estimates of M^t in each period (Fig. 13) are, on average, 12.6% of TIM^t_{UWVE}.

We can look at Fig. 13 in two ways. On the one hand we have a set of elements each including 10 estimates. In each element, estimates that correspond to consecutive periods are connected by an edge. On the other hand, we can consider a set of estimates in each period. In this case, one can think of different criteria for combining these estimates into sets of 10, one per period. With these two options at hand, we considered different sets of estimates for M^t and applied the CPLEX MILP solver with two stopping criteria: a time limit of 72 h and a mixed integer optimality gap tolerance of 0.03.

Among all tested instances, we selected to present and discuss five instances, further denoted by Inst-1,…,Inst-5. Inst-1 corresponds to the case where we set TIM^t_{UWVE} (Table 3, line 3) as the estimates for M^t. Instances Inst-2, Inst-3 and Inst-4 were the ones that gave the best RoadFor solutions among all the cases tested. Regarding Inst-2, the estimates for M^t, in each period t, were determined by calculating the median of the estimates in each period. Then, for each period, we randomly selected one value, from among the values below the median. Respecting Inst-3, for each period $t \in T$, we choose the minimum estimate given

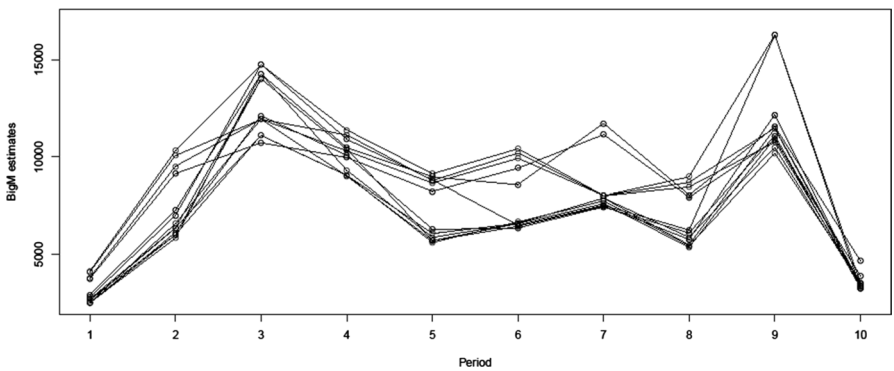

Fig. 13 Estimates for $M^t, t \in T$, given by 18 iterations of the Estimation procedure

Table 4 Dimensions of the reduced MILP after the execution of CPLEX MIP Presolve and aggregator

	# constraints	# variables	# binary var	# nonzeros
Inst-1,2,3,5	551015	582817	244830	1888726
Inst-4	469618	443770	216866	1516857

Table 5 Computational results for Inst-1 to Inst-5

Instance	Best obj (€)[1]	CL (€)[2]	CT (€)[3]	CB (€)[4]	CM (€)[5]	#feas sol[6]
Inst-1	2,098,522.9	410,858.2	**998,987.0**	357,443.6	331,234.2	197
Inst-2	2,097,945.3	411,255.8	1,000,977.2	352,393.0	333,319.2	56
Inst-3	2,097,878.8	410,934.9	999,375.9	359,312.0	328,256.0	172
Inst-4	**2,089,450.6**	410,907.1	1,002,548.4	**348,227.2**	**327,767.9**	137
Inst-5*	2,102,207.7	**409,893.4**	999,099.5	359,451.0	333,763.7	100

[1]Cost, in euros, of the best feasible solution obtained, calculated according to the MILP model objective function (1); [2]Sum of loading costs; [3]Sum of timber transportation costs; [4]Sum of road building costs; [5]Sum of road maintenance costs; [6]Number of feasible solutions found by CPLEX. Inst-5* terminated with CPLEX message "out of memory"

by the Estimation procedure (Fig. 13). As to Inst-4 we considered the same set of estimates as in Inst-3 but we extended the GraphSimplification procedure to remove all vertices, i $\notin VE$, with degree two covering the boundary of management units u $\in UVWE$. This option reduces the MILP problem dimensions in terms of vertices, edges and arcs, but it also restricts the choice of loading sites. The purpose of Inst-5 was to significantly tighten the value of M^t based on information given by previous RoadFor solutions. Thus, we considered the RoadFor solutions given by Inst-1 to Inst-4. For each solution s, s = 1,…,4, we determined the maximum timber flow along an arc, in each period $t \in T$, and denoted it by F_s^t. Then, we assigned to the estimate of M^t the minimum of the values obtained in period $t \in T$, $\min_s\{F_s^t\}$.

Before starting the branch&bound, CPLEX preprocesses the MILP problem, executing CPLEX MIP Presolve and Aggregator, in order to reduce its overall size. Table 4 displays the information given by CPLEX concerning the dimensions of the resulting reduced MILP problem. Row "Inst-1,2,3,5" presents average values.

Table 5 shows the best feasible solutions for the RoadFor problem given by CPLEX, for instances Inst-1 to Inst-5. The first column identifies the test instance. Column "Best obj (€)" shows the cost, in euros, of the best solution obtained, calculated according to the MILP model objective function (1). The following four columns display the costs associated with each component of (1). In particular, Column "CL (€)" displays the sum of loading costs, column "CT (€)" refers to the sum of timber transportation costs, column "CB (€)" gives the sum of road building costs and column "CM (€)" provides the sum of road maintenance costs. The best result obtained regarding "Best obj (€)" and each cost component is highlighted in bold. The last column presents the number of feasible solutions found by CPLEX. Note that, Inst-5 terminated with CPLEX ERROR 1001 – Out of memory solution. However, in such cases CPLEX displays the best solution saved.

3.2 Discussion

We can see that, along the planning horizon, the estimates provided by the Estimation procedure (Fig. 13) follow the same pattern of variability as the total amount of timber harvested in management units without exit road vertices in their boundaries, and in a slightly weaker way the variability of the total amount of timber harvested in all MUs. This becomes easier to see if we translate the values presented in the Table 3 to the barplot in Fig. 14 and compare it against Fig. 13. This comparison suggests that the reduction in timber flow capacities, based on information given by clusters of MUs, was able to capture the variability of the total timber flow along the planning horizon.

The computational results presented in Table 5 show that improvements in the objective function values are driven by reductions in road building and maintenance costs at the expense of an increase in timber loading and transportation costs. Balancing loading and transportation costs with build and maintenance costs lead to the use of longer routes for timber flow, in order to share roads, thus saving on the forest road network construction and maintenance costs.

The two best solutions for the RoadFor problem were obtained for instance Inst-3 and Inst-4. In both cases, the estimates of M^t correspond to the minimum values obtained by the Estimation procedure, in each period. Fig. 13 shows that these minimum values were reached in multiple iterations of the Heuristic algorithm. Moreover, the best solution was attained by Inst-4 whose underlying graph results from an extra simplification of the graph of instance Inst-3. The extra graph simplification in Inst-4 makes the corresponding MILP problem easier to solve allowing CPLEX to reach a better feasible solution. Moreover, it suggests that, in a future work, one may only consider vertices with degree greater than 2 and assign to each vertex in VP a loading cost that represents the loading costs in its neighborhood.

To evaluate the trade-off between building and maintenance decisions, Table 6 displays a description of Inst-4 solution in terms of the number of edges built and the number of edges maintained in each period. Results show that road construction

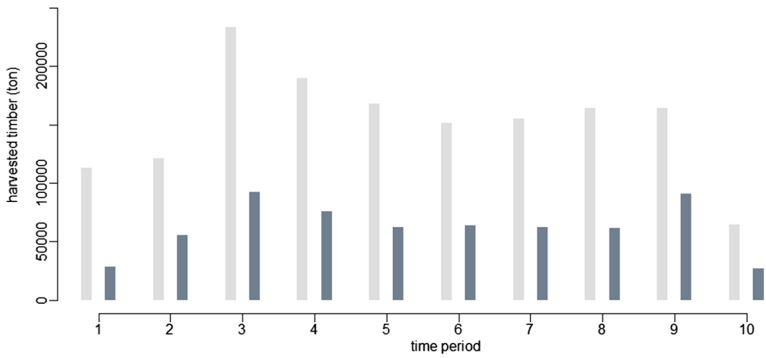

Fig. 14 Amount of timber harvested in each period (ton). Light grey bars represent the total amount in all MUs in each period. Dark grey bars show the total amount of timber harvested in MUs without exit road vertices at their boundaries

Table 6 Description of the best solution obtained in terms of the number of edges built, the number of edges maintained, in each period

Inst-4	Period $t \in T$									
	1	2	3	4	5	6	7	8	9	10
# edges built	576	620	195	30	3	18	11	15	11	11
# edges maintained	0	556	1161	1351	1359	1359	1357	1350	1360	821

takes place, mainly, in the first 3 years highlighting the need to provide the first access to many MUs in the forested landscape. Moreover, 93.36% of the total road edges built took place in the first 3 years. As to road maintenance, we can see that 96.53% of the road edges built in period t = 1, were maintained in period t = 2. The analysis for the next period shows that, 97.07% of the road edges built in periods t = 1 or t = 2, were maintained in period t = 3. Successively, we keep seeing that more than 93% of the edges built up to period t were kept in t + 1, with t = 1,…,8. There is a decrease in the number of edges maintained edges in period t = 10 that we believe to be related to a decrease in the amount of timber harvested in period t = 10.

Moreover, concerning Inst-4 solution, one grouped the edges used for timber transportation according to two criteria: (1) the number of periods an edge is built; (2) the number of periods an edge is maintained (Table 7).

One can see that the number of periods an edge was built from scratch attained the maximum value for k = 4 and this situation occurred for a single edge. Most of the edges used in the solution were built once (column k = 1). Concerning the decision to perform maintenance operations, we noticed that the maximum number of maintenance operations over an edge was attained for 573 edges, with each edge having been maintained in 8 periods of the planning horizon. That is, for the case study, most edges used in the solution for timber transportation were built once and were maintained over several periods, mostly 8 and 9 periods, highlighting the importance of integrating the building and maintenance decisions. Moreover, by combining the information in Tables 6 and 7 we see that we can see 61% of the edges built in the period t = 1 (Table 6) were maintained in the remaining k = 9 periods (Table 7).

Given the expected revenue of the input harvesting plan, 19,396 $\times 10^3$ euros, we can obtain the revenue of a plan encompassing both decisions on harvest scheduling,

Table 7 Description of the best solution obtained in terms of the number of edges built and the number of edges maintained in k periods

Inst-4	k periods								
# periods	k = 1	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
# edges built	1374	35	14	1	–	–	–	–	–
# edges maintained	9	8	19	23	23	111	281	573	352

Table 8 Cost, in euros, of the best feasible solutions given by CPLEX within the time limits of 6 h, 12 h and 24 h and the relative gap calculated as (|Best obj(Table 5) – Best obj(Table 8)|)/ Best obj(Table 5)

Instance	6 h cpu		12 h cpu		24 h cpu	
	Best obj_6 (€)	gap_6_72 (%)	Best obj_12 (€)	gap_12_72 (%)	Best obj_24 (€)	gap_24_72 (%)
Inst-1	2,156,637.7	2.8	2,129,084.2	1.5	2,111,407.7	0.6
Inst-2	2,400,245.9	14.4	2,141,369.2	2.1	2,102,674.5	0.2
Inst-3	2,399,650.0	14.4	2,153,135.7	2.6	2,105,654.2	0.4
Inst-4	2,400,245.9	14.9	2,134,928.6	2.2	2,096,083.2	0.3
Inst-5	2,382,866.3	13.4	2,133,148.8	1.5	2,108,052.0	0.3

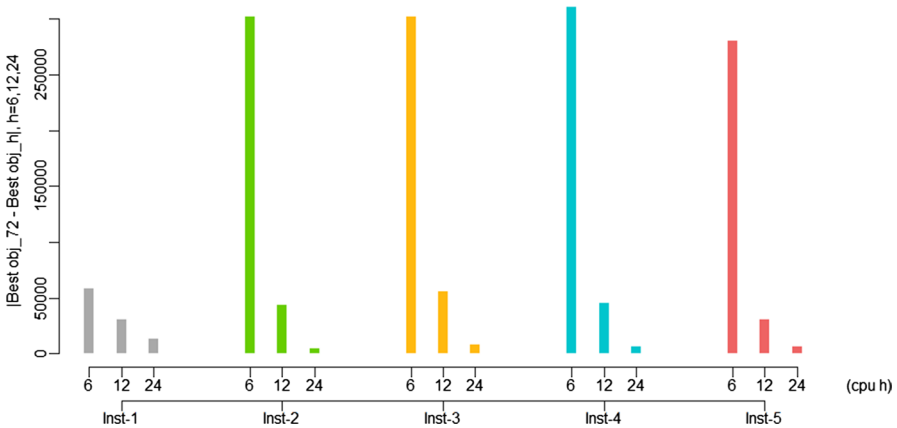

Fig. 15 For each instance, absolute gap between the cost in euros of the best solution given by CPLEX within the time limits of 6 h, 12 h, 24 h and the cost in euros of best solution given by CPLEX within a time limit of 72 h

forest road design and timber transportation by subtracting from the harvesting revenue the cost of the Inst-4 solution, 2096×10^3 euros.

Inst-5 has the lowest M^t estimate values, which were based on the values of timber flow given by the solutions of Inst-1,…,Inst-4. CPLEX terminated before reaching the time limit or the MIP tolerance due to an out of memory state. In this case, we think this is the result of having estimated M by very low values. However, for some other instances, whit M^t estimates between those of Inst-5 and those of Inst-1, CPLEX also ended prematurely due to the same error highlighting the difficulty to obtain "good" estimates for M^t.

Another question we may raise is whether it is worth waiting 72 h for a solution. The analysis of the best feasible solutions for the RoadFor problem obtained by CPLEX within a time limit of 6 h, 12 h and 24 h (Table 8) may help assess the impact of setting different CPLEX time limits.

For each instance, we also calculated the absolute gaps, according to (|Best obj_72 (Table 5) – Best obj_h (Table 8)|, h = 6, 12, 24 (Fig. 15).

Table 8 and Fig. 15 suggest that the CPLEX time limit should be greater than or equal to 24 h, so that solutions of good quality are achieved. In particular, for the time limit of 24 h, we obtained 12,884.8(€), 4729.2(€), 7775.44(€), 6632.6(€) and 5844.3(€), for Inst-1,…,Inst-5, respectively. Dividing these gaps by the 7623 ha, area of the forested landscape under study, one obtains the residual values of 1.69(€/ha), 0.62(€/ha), 1.02(€/ha), 0.87(€/ha) and 0.77(€/ha), in the planning horizon of 100 years, respectively for Inst-1,…,Inst-5.

The current work was motivated by a real-life case. Applying mathematical models to solve real life forest management problems requires the compilation of information from landowners and expert forest planners. Based on this information we developed the mathematical model presented in Sect. 2.2.1. The underlying objective function includes 4 components representing the main costs to be optimized. Table 5 shows the value that each of these components takes for the best solutions obtained. We may notice that the greatest contributions are made by the two components related to timber transportation and timber loading costs. This may be explained by the fact that these contributions depend on the total timber harvested in each period. As to the loading cost magnitude, additionally to the timber loading cost, it takes into consideration the suitability of that location to serve as the access point to the management unit, not only for the vehicles that transport timber out of the forest, but also for the machinery that performs harvesting operations.

We decided to allow segment roads to be built and, or, maintained more than once, depending on what is most advantageous in terms of cost. Our option increases the computational complexity of the problem to be solved. In alternative, while dealing with the integrated harvesting and road design problem, Andalaft et al. (2003) and Naderializadeh and Crowe (2018) considered that road construction occurs only once over the planning horizon while Ross et al. (2018) and Bellavenutte et al. (2020) assume that the road network already exists and can be maintained whenever advisable for timber transportation.

To solve the forest road building or maintenance and timber transportation problem, suggested by the forest owners from the joint collaborative management area – ZIF Paiva, we developed a new Estimation procedure to tighten big-M constraints (4) and (5). Andalaft et al. (2003) also recognized the importance of reducing the big-M value. They reduced the capacity of the arcs to the maximum amount of timber that can flow in the arc. For tree like road networks this procedure is able to improve the solutions quality. However, as the number of cycles in the network increases the effectiveness of this strategy decreases. For the case of road networks with cycles, Naderializadeh and Crowe (2018)) reported their unsuccessful experience of replacing big-M constraints with "indicator constraints" available in CPLEX 12.5. The big-M Estimation procedure proposed in Sect. 2.2.2.2 of the current paper can be applied to any type of road network. Moreover, computational results showed that it was able to improve the quality of the feasible solutions obtained.

The Estimation procedure is embedded in an iterative scheme that relies on the execution of a clustering algorithm. The set of MUs is partitioned into clusters based on identification of shortest distance routes to exit road vertices that may be potentially shared by a group of MUs. Estimates for $M^t, t \in T$ are obtained taking into account the total amount of timber harvested in each cluster and in each period. The

resulting MILP model, embracing the set of all MUs and all periods of the planning horizon, is then solved by a heuristic branch-and-bound.

An iterative heuristic approach based on a clustering algorithm has already been proposed by Bellavenutte et al. (2020) to address an integrated harvest scheduling and road maintenance problem. These authors proposed a k-means algorithm that divides the forest into clusters of MUs based on the shortest road distance between MUs. Cluster subproblems are then solved independently. This partitioning approach is embedded in an iterative procedure, that swapes timber volume production targets between subproblems, aiming to improve the solution for the integrated problem.

4 Conclusions and future work

Planning forest roads and timber transportation can be a challenging problem due to its size and complexity. We considered a real study case with 1,137 management units, 10,785 potential timber loading sites and 13,086 potential road segments. The planning horizon was extended over ten 10-year periods. First, we used a GIS based approach to develop a representation of the problem, which was modeled within a Mixed Integer Programming (MIP) framework. However, a straightforward use of a solver (CPLEX) for this MIP model was not able to provide solutions. Indeed, the instance was too large, in terms of the number of vertices, edges and arcs, and the Linear Programming (LP) relaxation bounds were too weak, making the solver procedure inefficient. Indeed, the dimension of the instance was too large, in terms of the number of vertices, edges and arcs, and the Linear Programming (LP) relaxation bounds were too weak, making the solver procedure inefficient.

The reason those bounds are poor is related with the so called big-M variable upper bound constraints in the formulation, which allow loose values for integer variables in the LP relaxation. Thus, to overcome this issue we proposed a procedure to estimate tighter variable upper bounds. The approach allowed CPLEX to obtain feasible solutions that were considered to be of good quality, by expert forest planners.

In a near future, we intend to study mathematical formulations alternative to the one presented in the current paper and to develop decomposition approaches to reduce the size of the problems to solve.

Acknowledgements This research has received funding from the BIOECOSYS project “Forest ecosystem management decision-making methods an integrated bio-economic approach to sustainability” [LISBOA-01-0145-FEDER-030391, PTDC/ASP-SIL/30391/2017)] and from the MODFIRE project "A multiple criteria approach to integrate wildfire behavior in forest management planning (PCIF/MOS/0217/2017). The authors would like to thank Fundação para a Ciência e a Tecnologia I.P. (FCT), Portugal, for funding through the Center of Mathematics, Fundamental Applications and Operations Research Project UIDB/04561/2020 and through the Forest Research Centre Project UIDB/00239/2020; through the Ph.D. grants of Marlene Marques (PD/BD/128257/2016) and Marco Marto (SFRH/BD/108225/2015) and the contract in the scope of Norma Transitória – DL57/2016/CP1382/CT15 of Susete Marques. The authors thank the anonymous reviewers for their helpful comments.

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